Unit 1: Structure, Bonding and Introduction to Organic Chemistry

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit gives students opportunities to develop the basic chemical skills of writing formulae and equations, and calculating chemical quantities.

The study of atomic structure includes a description of *s*, *p*, and *d* orbitals and shows how electronic configurations can account for the arrangement of elements in the Periodic Table. This leads to an appreciation of one of the central features of chemistry: the explanation of the properties of elements and the patterns in the Periodic Table in terms of atomic structure.

An understanding of the electronic structure of atoms leads to an appreciation of the three types of strong chemical bonding: ionic, covalent and metallic. Following from this, shapes of molecules can then be considered.

The basic principles of organic chemistry are covered and students study alkanes and alkenes, and will begin to develop a mechanistic approach to organic chemistry.

Chemistry in action

The study of atomic structure gives some insight into the methods that scientists use to study the structure of atoms. This leads to the introduction of the mass spectrometer and its importance in sensitive methods of analysis in areas such as space research, medical research and diagnosis, in detecting drugs in sport and in environmental monitoring.

Chemists set up theoretical models and gain insight by comparing real and theoretical properties of chemicals. This is illustrated in the unit by considering the evidence for the different kinds of chemical bonding.

Electron-pair repulsion theory is also used to show how chemists can develop theories and use them to make predictions.

Students start to use the conventions for mechanisms in organic chemistry as a way to represent the movement of electrons in reactions.

Practical skills	Students can start with simple test-tube reactions to illustrate a range of chemical equations. They can then build up to carrying out practical work that can be used to find reaction quantities, covered in the first core practical on molar volume. Simple practical work can be used to investigate the properties of substances with different types of bonding. The introduction to organic chemistry shows how chemists work safely with hazardous chemicals by managing risks. A number of possible practicals can be used to explore the chemistry of alkenes.
Mathematical skills	There are opportunities for the development of mathematical skills in this unit. This includes converting between units such as cm^3 and dm^3, using standard form with the Avogadro constant, rearranging formulae for calculating moles in solids and in solutions and the ideal gas equation, calculating atom economy, dealing with percentage errors, calculating a relative atomic mass from isotopic composition data, using simple probability to calculate the peak heights for the mass spectrum of molecules such as chlorine, using logarithms to compare successive ionisation energies for an element, representing shapes of molecules with suitable sketches, plotting data to investigate trends in boiling temperatures of alkanes and calculating the yield of a reaction. (Please see *Appendix 6: Mathematical skills and exemplifications* for more detail.)

Assessment information

- First assessment: January 2019.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 80 marks.
- Students must answer all questions.
- This paper has two sections:
  - Section A: multiple choice questions
  - Section B: mixture of short-open, open-response and calculation questions.
- This paper will include a minimum of 18 marks that target mathematics at Level 2 or above.
- Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
- Calculators may be used in the examination (see *Appendix 8: Use of calculators*).

Topic 1: Formulae, Equations and Amount of Substance

Application of ideas from this topic will be applied to all other units.

Students will be assessed on their ability to:

1.1	know the terms 'atom', 'element', 'ion', 'molecule', 'compound', 'empirical formula' and 'molecular formula'
1.2	know that the mole (mol) is the unit for the amount of a substance and be able to perform calculations using the Avogadro constant *L* (6.02×10^{23} mol^{-1})
1.3	write balanced full and ionic equations, including state symbols, for chemical reactions
1.4	understand the terms: i 'relative atomic mass' based on the ^{12}C scale ii 'relative molecular mass' and 'relative formula mass', including calculating these values from relative atomic masses *The term 'relative formula mass' should be used for compounds with giant structures.* iii 'molar mass' as the mass per mole of a substance in g mol^{-1} iv parts per million (ppm), including gases in the atmosphere
1.5	calculate the concentration of a solution in mol dm^{-3} and g dm^{-3} *Titration calculations are not required at this stage.*
1.6	be able to use experimental data to calculate empirical and molecular formulae
1.7	be able to use chemical equations to calculate reacting masses and vice versa, using the concepts of amount of substance and molar mass
1.8	be able to use chemical equations to calculate volumes of gases and vice versa, using: i the concepts of amount of substance ii the molar volume of gases iii the expression $pV = nRT$ for gases and volatile liquids
1.9	be able to calculate percentage yields and percentage atom economies (by mass) in laboratory and industrial processes, using chemical equations and experimental results Atom economy = $\frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all products}} \times 100\%$
1.10	be able to determine a formula or confirm an equation by experiment, including evaluation of the data
1.11	**CORE PRACTICAL 1** **Measurement of the molar volume of a gas.**
1.12	be able to relate ionic and full equations, with state symbols, to observations from simple test-tube experiments, to include: i displacement reactions ii typical reactions of acids iii precipitation reactions

	Further suggested practicals: i preparation of a salt and calculating the percentage yield of product, including the preparation of a double salt, such as ammonium iron(II) sulfate from iron, ammonia and sulfuric acid ii determine a chemical formula by experiment, such as the formula of copper(II) oxide by reduction iii determine a chemical equation by experiment, such as the reaction between lithium and water, or the reaction between magnesium and an acid iv carry out and interpret the results of simple test-tube reactions, as outlined in 1.12